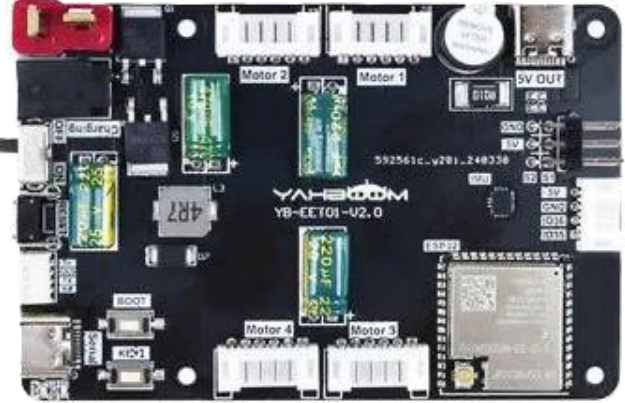


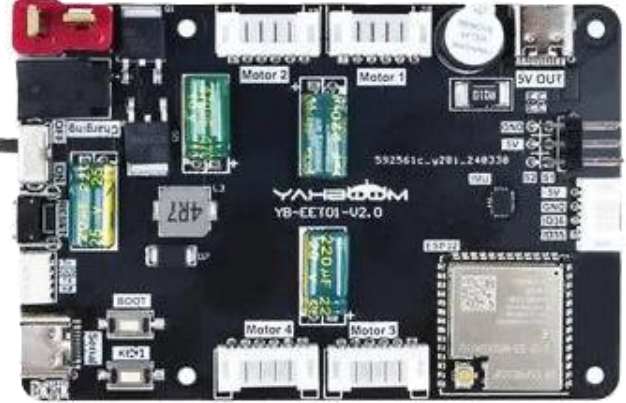
1 Introduction

This group project aims to provide students with hands-on experience in designing and fabricating a smart AI wheeled logistics robot that integrates robotic systems with a Vision AI system, employing 3D printing and laser material processing. Each group's smart robot should be equipped with an electric lifter, which allows it to carry and relocate a rack from one position to another. Each group will be provided with the following components:

Item	Description	QTY	Remark
1	Micro-ROS Controller Board for Raspberry Pi 5 	1	- Running micro-ROS firmware. - Controlling sensors and actuators. - Providing power to the Raspberry Pi 5. - https://www.yahboom.net/study/MicroROS-Board

1 引言

本小组项目旨在让学生获得设计与制作智能 AI 轮式物流机器人的动手实践经验。该机器人将机器人系统与视觉 AI（Vision AI）系统相集成，并运用 3D 打印和激光材料加工（laser material processing）技术制作。每组的智能机器人应配备电动升降装置（electric lifter），使其能够承载货架（rack）并将其从一个位置搬运到另一个位置。每组将获得以下元器件：

序号	描述	数量	备注
1	适用于 Raspberry Pi 5 的 Micro-ROS 控制板 	1	- 运行 micro-ROS 固件。 - 控制传感器和执行器（actuator）。 - 为 Raspberry Pi 5 供电。 - https://www.yahboom.net/study/MicroROS-Board

Item	Description	QTY	Remark
2	USB Type A to Type C cable (for Pi 5 data) 	1	- Connect the micro-ROS controller board with the Raspberry Pi 5. - For USB-Serial communication
3	USB Type C to Type C cable (for Pi 5 power) 	1	- Connect the micro-ROS controller board with the Raspberry Pi 5. - For powering the Raspberry Pi 5 with a battery.

序号	描述	数量	备注
2	USB Type A 转 Type C 线缆（用于 Pi 5 数据传输） 	1	- 连接 micro-ROS 控制板与 Raspberry Pi 5。 - 用于 USB 串口（USB-Serial）通信
3	USB Type C 转 Type C 线缆（用于 Pi 5 供电） 	1	- 连接 micro-ROS 控制板与 Raspberry Pi 5。 - 用于通过电池为 Raspberry Pi 5 供电。

Item	Description	QTY	Remark
4	Raspberry Pi 5 8GB Board with heatsink and fan 	1	- Running Linux and ROS2. - Controlling the camera and Lidar. - Wi-Fi networking. - https://www.raspberrypi.com/documentation/computers/getting-started.html
5	32GB Memory Card (for Raspberry Pi 5) 	1	- Installed on the Raspberry Pi 5. - Storage for operating systems and user data.
6	USB Card Reader	1	- Read and write a memory card on a computer.

序号	描述	数量	备注
4	Raspberry Pi 5 8GB 主板（含散热片和风扇） 	1	- 运行 Linux 和 ROS2。 - 控制摄像头和激光雷达（Lidar）。 - Wi-Fi 联网。 - https://www.raspberrypi.com/documentation/computers/getting-started.html
5	32GB 存储卡（用于 Raspberry Pi 5） 	1	- 安装在 Raspberry Pi 5 上。 - 用于存储操作系统和用户数据。
6	USB 读卡器	1	- 在电脑上读写存储卡。

Item	Description	QTY	Remark
7	Raspberry Pi Camera Module 3 with cable 	1	- Connect to the Raspberry Pi 5. - https://www.raspberrypi.com/documentation/accessories/camera.html
8	Raspberry Pi 5 Power Supply 	1	- Powering the Raspberry Pi temporarily during the development stages. - The smart AI wheeled logistics robot must be battery-powered.

序号	描述	数量	备注
7	Raspberry Pi Camera Module 3 摄像头模块（含排线） 	1	- 连接至 Raspberry Pi 5。 - https://www.raspberrypi.com/documentation/accessories/camera.html
8	Raspberry Pi 5 电源适配器 	1	- 在开发阶段临时为 Raspberry Pi 供电。 - 智能 AI 轮式物流机器人必须采用电池供电。

Item	Description	QTY	Remark
9	Raspberry Pi 5 Display Cable 	1	- Connect an external display (LCD) temporarily during the development stages.
10	MS200 Lidar with cable and adapter board  MS200 Laser Lidar  Serial adapter board  Connection cable	1	- Connect to Raspberry Pi 5 for map creation and autonomous navigation. - https://www.yahboom.net/study/MS200

序号	描述	数量	备注
9	Raspberry Pi 5 显示器线缆 	1	- 在开发阶段临时连接外部显示器（LCD）。
10	MS200 激光雷达（含线缆和转接板）  MS200 激光雷达  串口转接板  连接线	1	- 连接至 Raspberry Pi 5，用于地图构建和自主导航。 - https://www.yahboom.net/study/MS200

Item	Description	QTY	Remark
11	USB Type A to Type C cable (for Lidar data) 	1	- Connect the MS200 Lidar to the Raspberry Pi 5.
12	Mecanum wheels (Left) 65mm 	2	

序号	描述	数量	备注
11	USB Type A 转 Type C 线缆（用于激光雷达数据传输） 	1	- 将 MS200 激光雷达连接至 Raspberry Pi 5。
12	麦克纳姆轮（Mecanum wheel，左）65mm 	2	

Item	Description	QTY	Remark
13	Mecanum wheels (Right) 65mm 	2	
14	Mecanum wheels adapter for 310 motor with screws 	4	- Connect a 65mm Mecanum wheel with a 310 DC Encoder Motor.
15	310 DC Encoder Motor with cable 	4	- The white colour connector plugs into the micro-ROS controller board. - The black colour connector plugs into the 310 motor. - Make sure the connection is correct before powering on the system. - PPR: 13. Gear ratio: 1:20. - An incorrect connection will burn the controller board.

序号	描述	数量	备注
13	麦克纳姆轮（右）65mm 	2	
14	310 电机用麦克纳姆轮转接件（含螺丝） 	4	- 将 65mm 麦克纳姆轮与 310 直流编码电机相连。
15	310 直流编码电机（DC Encoder Motor，含线缆） 	4	- 白色接头插入 micro-ROS 控制板。 - 黑色接头插入 310 电机。 - 系统通电前，务必确认连接正确。 - PPR：13；减速比（Gear ratio）：1:20。 - 接线错误将烧毁控制板。

Item	Description	QTY	Remark
16	7.4v Lithium battery 9900mAh with charger 	1	- Do not short the power leads. - Do not overcharge the battery. - Do not charge the battery unattended.
17	Battery Charger 	1	
18	CN to HK power plug adapter 	1	

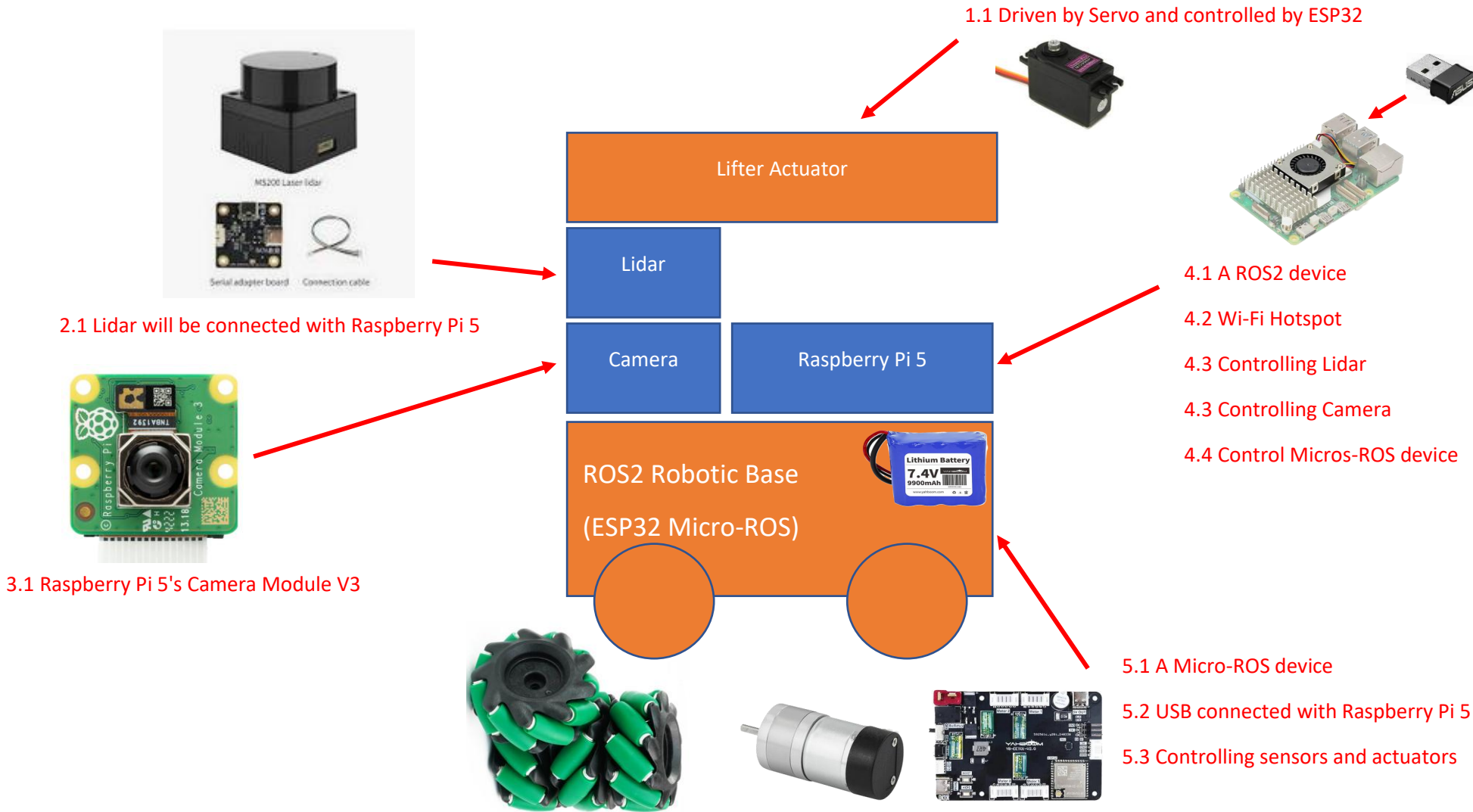
序号	描述	数量	备注
16	7.4v 锂电池 9900mAh（含充电器） 	1	- 切勿使电源引线短路。 - 切勿对电池过度充电。 - 切勿在无人看管时为电池充电。
17	电池充电器 	1	
18	CN 转 HK 电源插头转换器（内地转香港） 	1	

Item	Description	QTY	Remark
19	MG996 Servo (180 degree) 	1	- For moving a platform up and down. - Wire: Black (GND), RED (+5V), Yellow (PWM).
20	USB Wi-Fi Dongle 	1	- For creating a Hotspot on the Raspberry Pi 5

序号	描述	数量	备注
19	MG996 舵机 (Servo , 180 度) 	1	- 用于带动平台上下移动。 - 接线：黑色 (GND) ，红色 (+5V) ，黄色 (PWM) 。
20	USB Wi-Fi 无线网卡 	1	- 用于在 Raspberry Pi 5 上创建热点 (Hotspot)

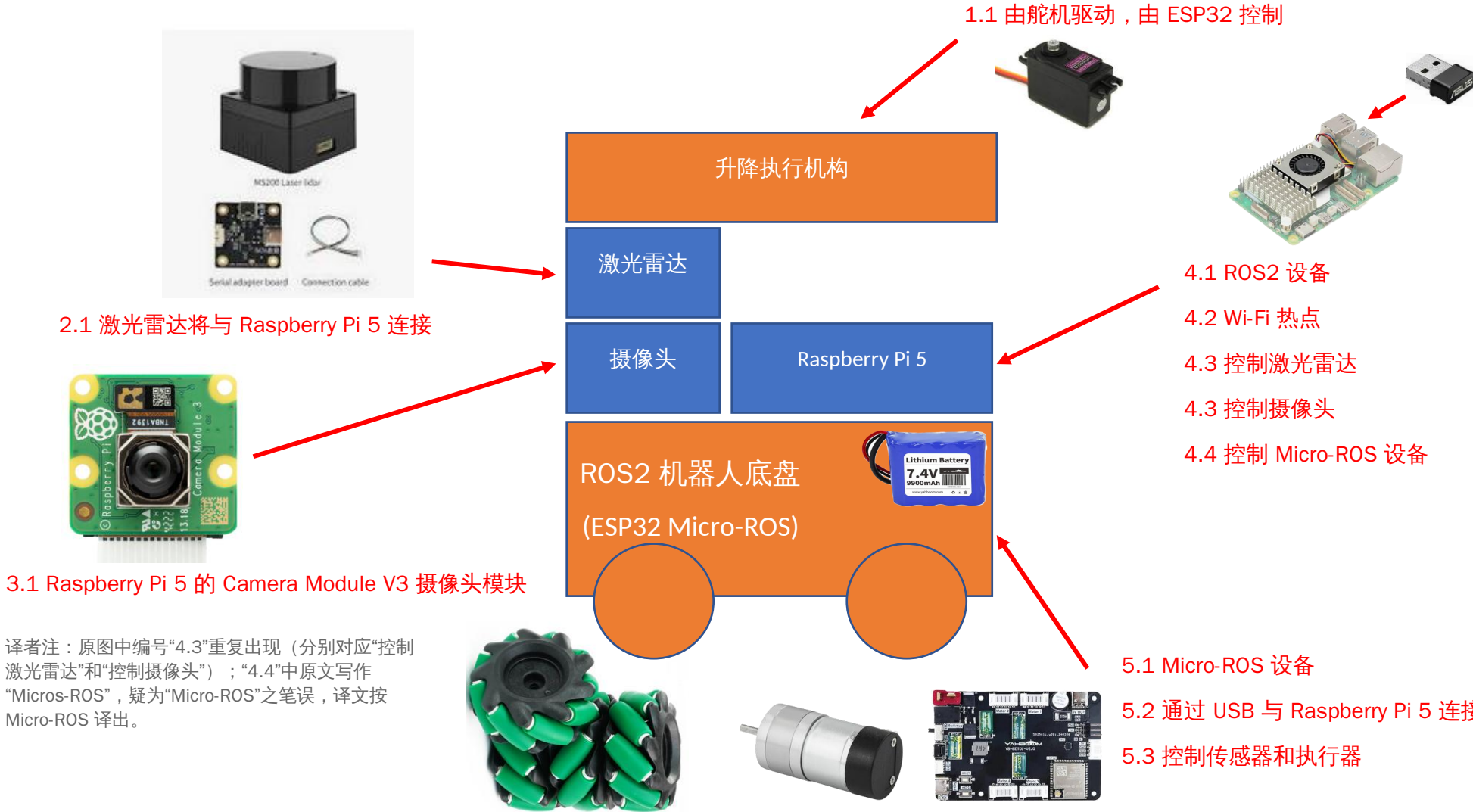
2 A simplified block diagram showing the possible building blocks of a robot

This block diagram is an example only. Each group should have their own unique robot design.



2 机器人可能组成模块的简化框图

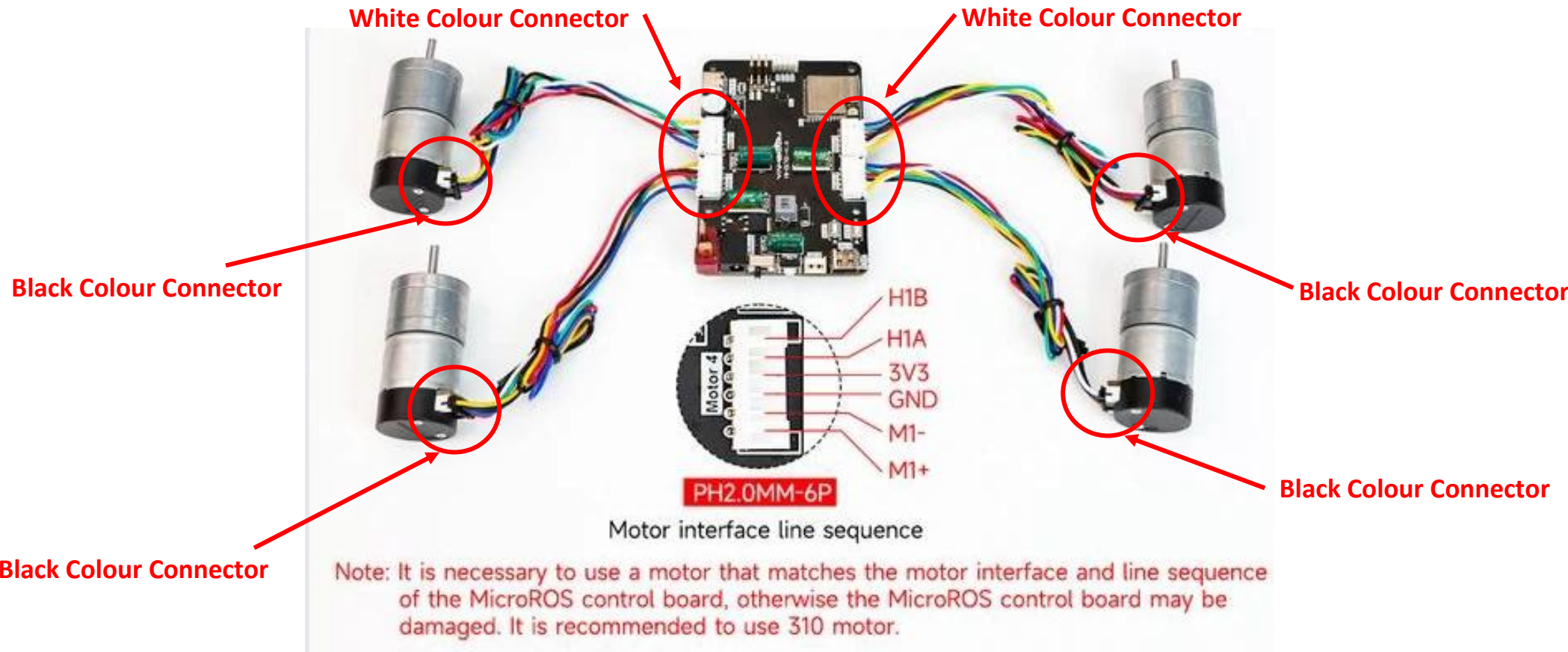
本框图仅为示例。各小组应有各自独特的机器人设计。



译者注：原图中编号“4.3”重复出现（分别对应“控制激光雷达”和“控制摄像头”）；“4.4”中原文写作“Micros-ROS”，疑为“Micro-ROS”之笔误，译文按 Micro-ROS 译出。

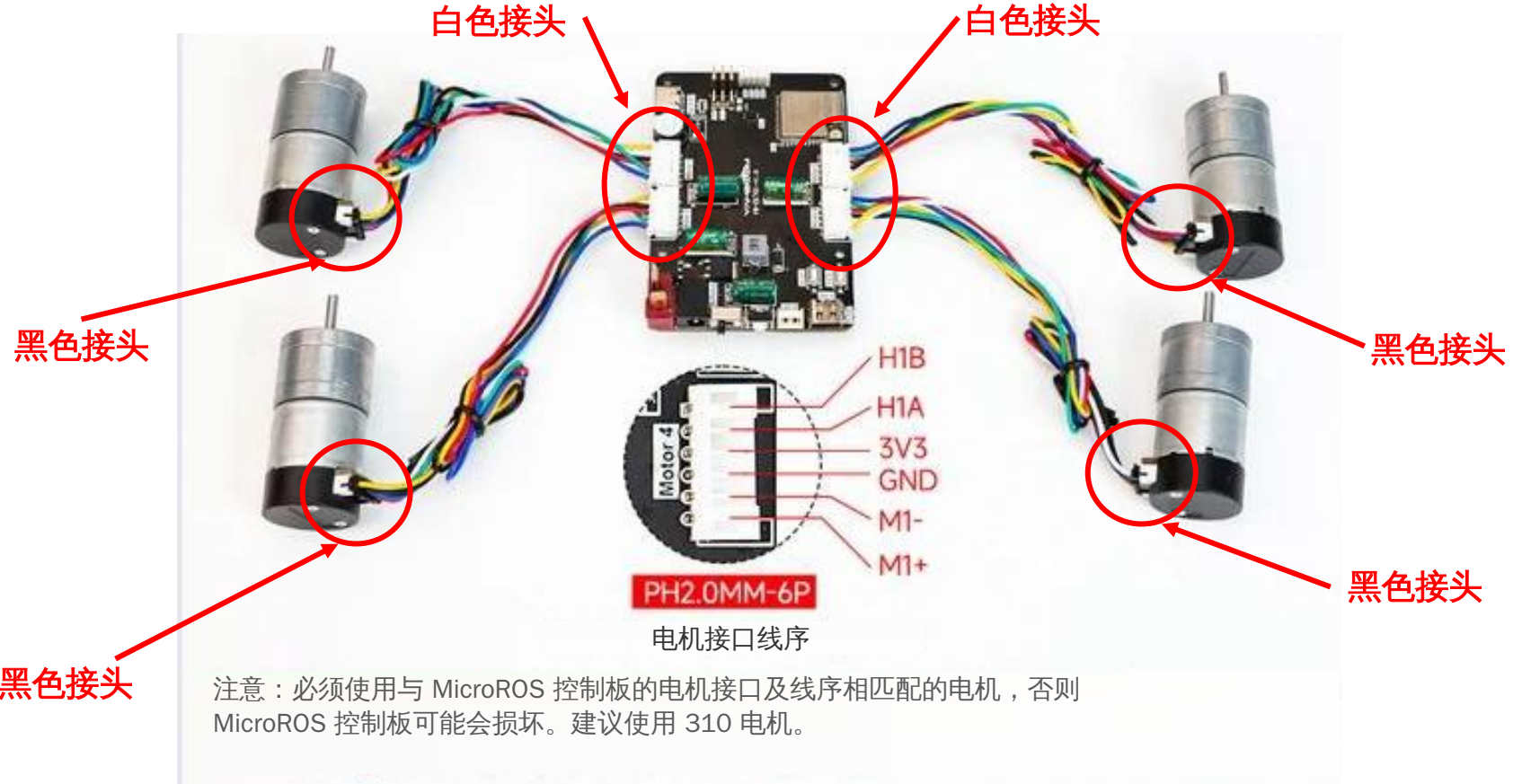
3 A connection diagram of the Micro-ROS Controller Board with 310 Encoder Motors.

Before powering up the system, double-check that all motor cables are installed in the correct order by verifying the connector colour. The motor cable has two connectors. The white colour connector is connected to the Micro-ROS controller board. The black colour connector is connected to the motor. Incorrect installation of motor cables may cause damage to the Micro-ROS controller board.



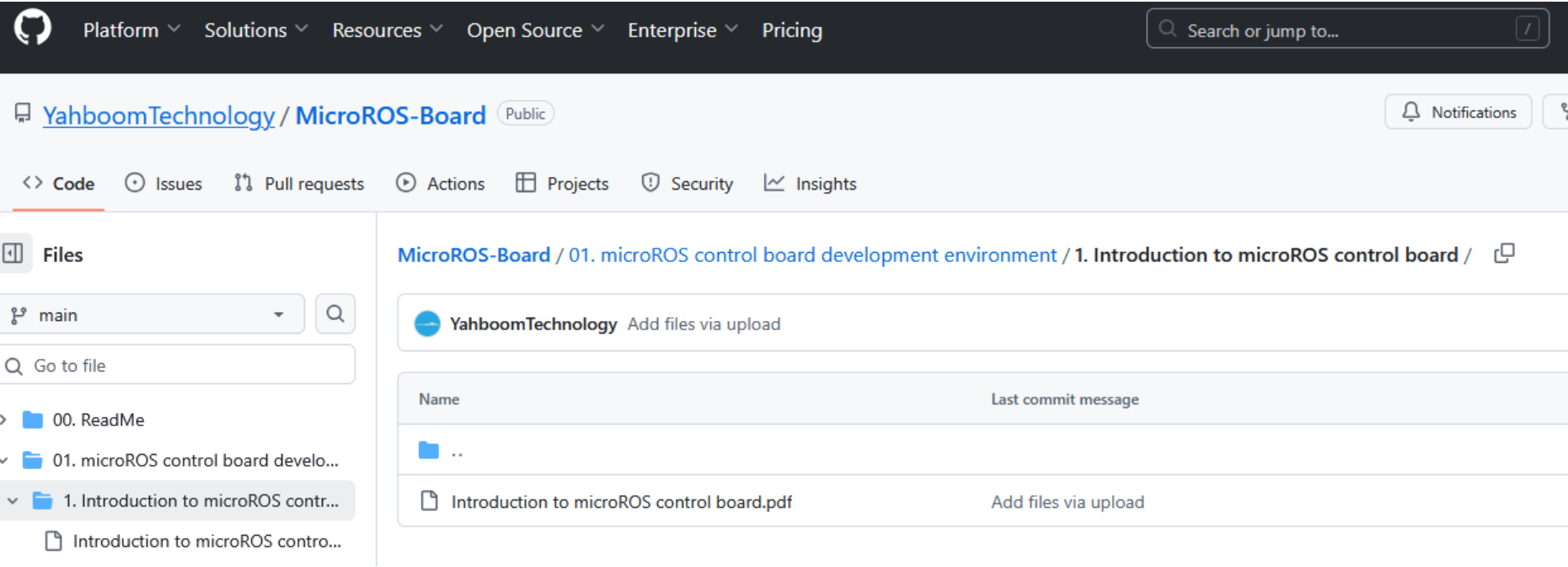
3 Micro-ROS 控制板与 310 编码电机的连接示意图

在系统上电前，请通过核对接头颜色，再次确认所有电机线缆均已按正确顺序安装。电机线缆有两个接头。白色接头连接 Micro-ROS 控制板，黑色接头连接电机。电机线缆安装错误可能导致 Micro-ROS 控制板损坏。



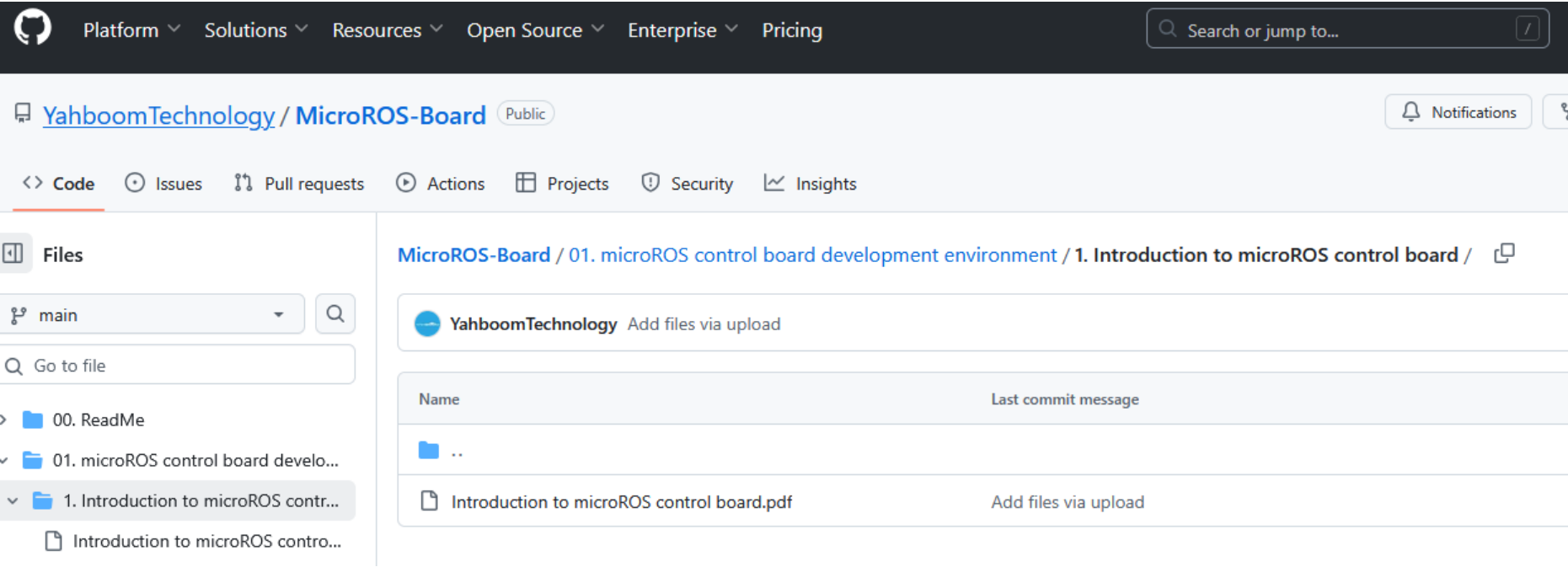
4 Hardware Resources of the Micro-ROS Controller Board

The vendor’s GitHub website (<https://github.com/YahboomTechnology/MicroROS-Board>) provides details about the hardware of the controller board. The location is “01. microROS control board development environment/1. Introduction to microROS control board”.

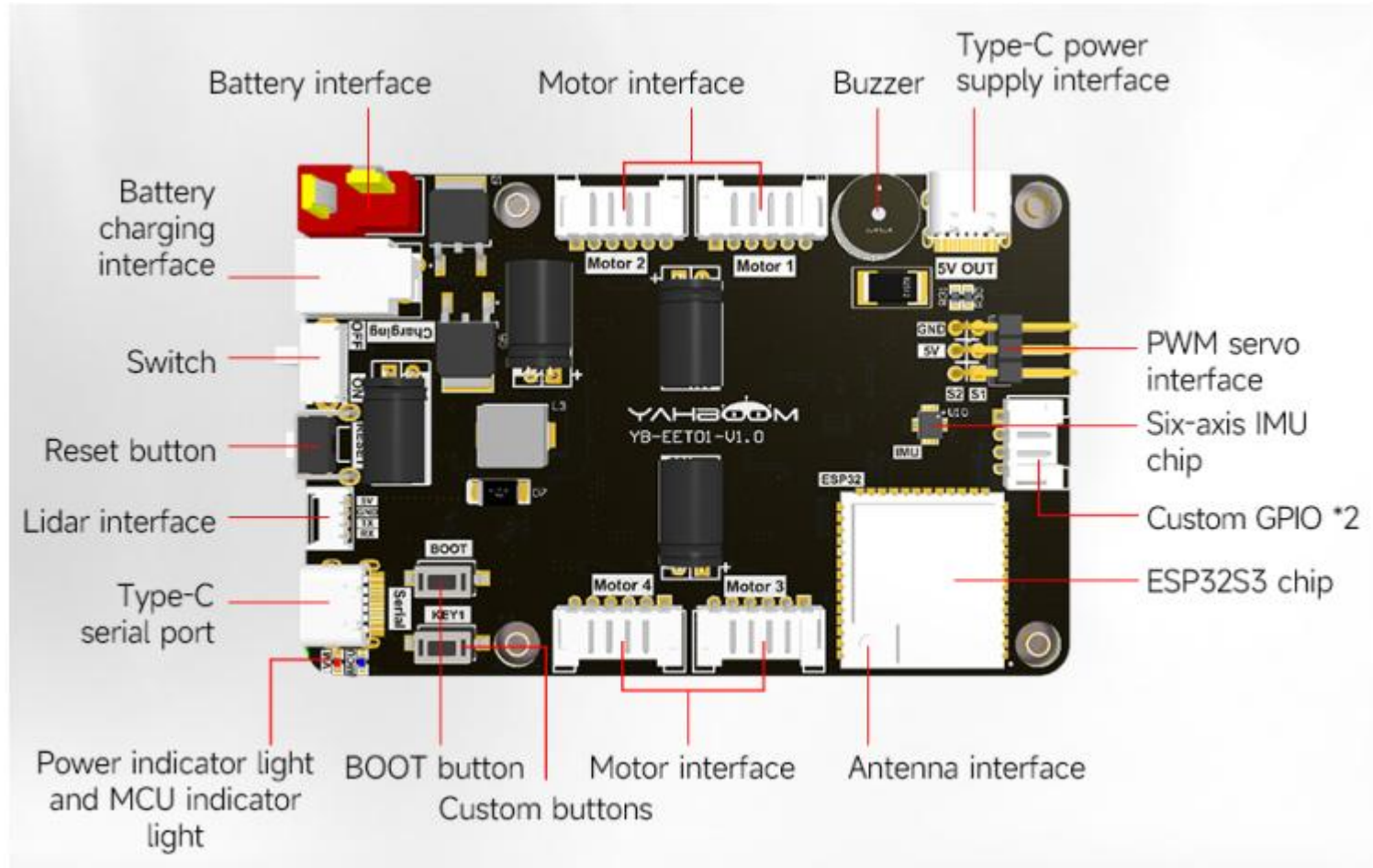


4 Micro-ROS 控制板的硬件资源

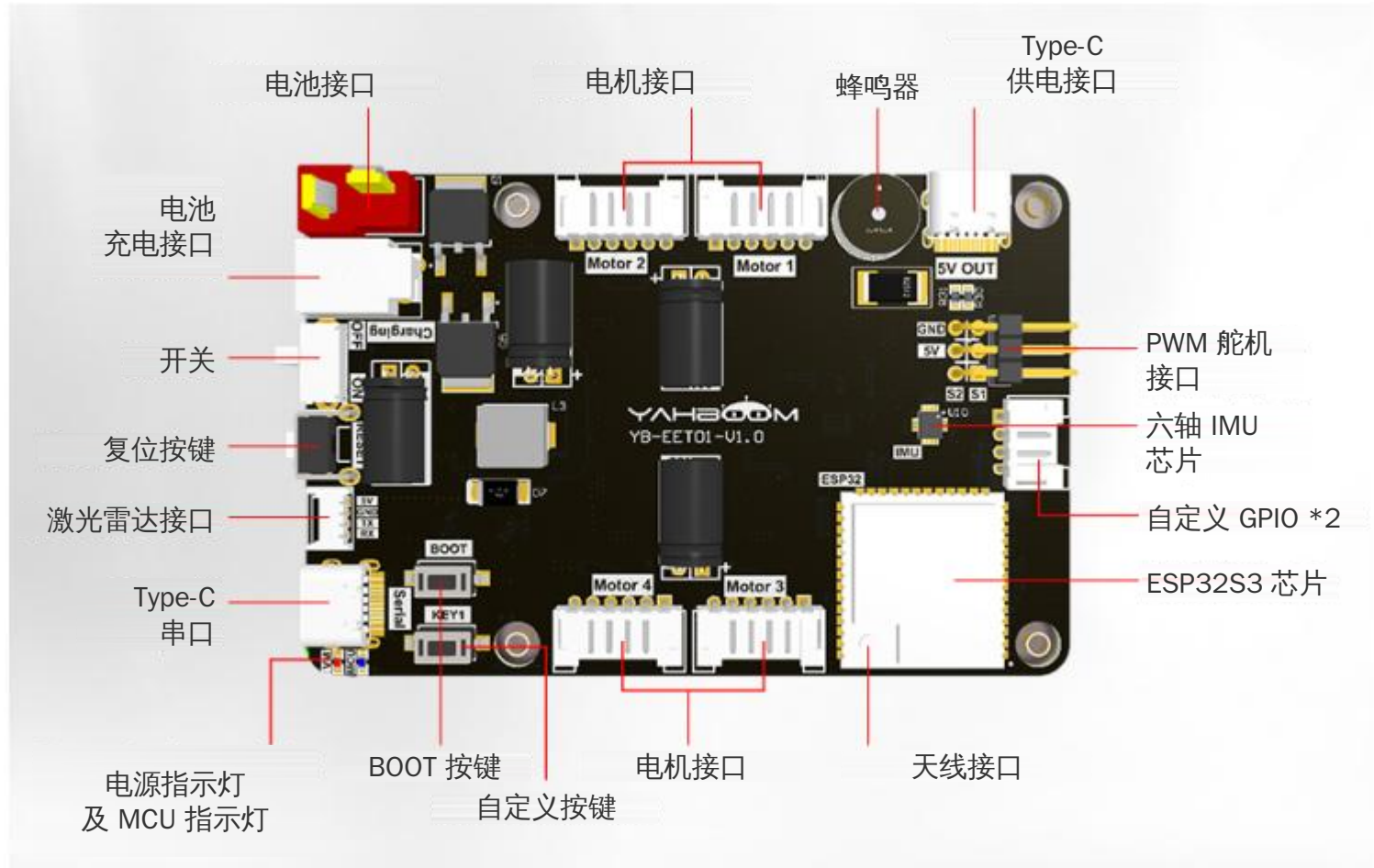
供应商的 GitHub 网站 (<https://github.com/YahboomTechnology/MicroROS-Board>) 提供了该控制板的硬件详细信息，位置为“01. microROS control board development environment/1. Introduction to microROS control board”（即“01. microROS 控制板开发环境/1. microROS 控制板介绍”）。



For example, there is a diagram that describes the available hardware resources.



例如，其中有一张描述可用硬件资源的示意图。



The document also provides a pin assignment table.

Name of peripheral device	ESP32-S3-GPIO
Motor M1-PWM-M1A	GPIO4
MotorM1-PWM-M1B	GPIO5
MotorM1-encoder-H1A	GPIO6
MotorM1-encoder-H1B	GPIO7
MotorM2-PWM-M2A	GPIO15
MotorM2-PWM-M2B	GPIO16
MotorM2-encoder-H2A	GPIO47
MotorM2-encoder-H2B	GPIO48
MotorM3-PWM-M3A	GPIO9
MotorM3-PWM-M3B	GPIO10
MotorM3-encoder-H3A	GPIO11
MotorM3-encoder-H3B	GPIO12
MotorM4-PWM-M4A	GPIO13
MotorM4-PWM-M4B	GPIO14
MotorM4-encoder-H4A	GPIO1
MotorM4-encoder-H4B	GPIO2
BOOT button-IO0	GPIO0
Custom key-KEY1	GPIO42
Battery voltage detection-BAT	GPIO3
MCU indicator light-LED	GPIO45
Buzzer	GPIO46
Servo interface -S1	GPIO8
Servo interface -S2	GPIO21
IMU interrupts -INT	GPIO41
IMU-I2C-SCL	GPIO39
IMU-I2C-SDA	GPIO40
Radar RX-serial port 1-TXD1	GPIO17
Radar TX-serial port 1-RXD1	GPIO18
Type-C serial port RX-TXD0	GPIO43

Name of peripheral device	ESP32-S3-GPIO
Type-C serial port TX-RXD0	GPIO44
Custom GPIO	GPIO35
Custom GPIO	GPIO36
Custom indicator lights	D1
Custom indicator lights	D2

该文档还提供了一张引脚分配表。

外设名称	ESP32-S3-GPIO
电机 M1-PWM-M1A	GPIO4
电机 M1-PWM-M1B	GPIO5
电机 M1-编码器-H1A	GPIO6
电机 M1-编码器-H1B	GPIO7
电机 M2-PWM-M2A	GPIO15
电机 M2-PWM-M2B	GPIO16
电机 M2-编码器-H2A	GPIO47
电机 M2-编码器-H2B	GPIO48
电机 M3-PWM-M3A	GPIO9
电机 M3-PWM-M3B	GPIO10
电机 M3-编码器-H3A	GPIO11
电机 M3-编码器-H3B	GPIO12
电机 M4-PWM-M4A	GPIO13
电机 M4-PWM-M4B	GPIO14
电机 M4-编码器-H4A	GPIO1
电机 M4-编码器-H4B	GPIO2
BOOT 按键-IO0	GPIO0
自定义按键-KEY1	GPIO42
电池电压检测-BAT	GPIO3
MCU 指示灯-LED	GPIO45
蜂鸣器	GPIO46
舵机接口-S1	GPIO8
舵机接口-S2	GPIO21
IMU 中断-INT	GPIO41
IMU-I2C-SCL	GPIO39
IMU-I2C-SDA	GPIO40
雷达 RX-串口 1-TXD1	GPIO17
雷达 TX-串口 1-RXD1	GPIO18
Type-C 串口 RX-TXD0	GPIO43

外设名称	ESP32-S3-GPIO
Type-C 串口 TX-RXD0	GPIO44
自定义 GPIO	GPIO35
自定义 GPIO	GPIO36
自定义指示灯	D1
自定义指示灯	D2

5 ROS2 Node for the Raspberry Pi Camera Modules

The ROS2 package "camera_ros" supports Raspberry Pi Camera Modules, including Camera Module 3. More information is available on the Github page https://github.com/christianrauch/camera_ros. Both "libcamera" and "camera_ros" packages should be built from source code.

6 Acknowledgement

1. Vendor’s GitHub website for the Micro-ROS Controller Board.

URL: <https://github.com/YahboomTechnology/MicroROS-Board>

5 适用于 Raspberry Pi 摄像头模块的 ROS2 节点

ROS2 软件包“camera_ros”支持 Raspberry Pi 摄像头模块，包括 Camera Module 3。更多信息请参见其 GitHub 页面 https://github.com/christianrauch/camera_ros。“libcamera”和“camera_ros”两个软件包均应从源代码编译构建。

6 致谢

1. Micro-ROS 控制板供应商的 GitHub 网站。

URL: <https://github.com/YahboomTechnology/MicroROS-Board>